

Chapter 2: Gamma, Beta, Zeta

Mathematical Physics

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Where we are going

Many special functions begin as discrete objects—factorials, partial sums, p -series—and gain power once they are extended to continuous (or complex) arguments. This chapter develops the two functions that form the basic *language* of that extension:

- **Harmonic numbers** $H_n = \sum_{k=1}^n \frac{1}{k}$ and the continuous interpolation that produces the Euler–Mascheroni constant γ .
- The **gamma function** $\Gamma(z)$: the canonical meromorphic interpolation of the shifted factorial. Integral definition, recursion, half-integers, poles, reflection, Euler's limit form, and Stirling.
- The **beta function** $B(p, q)$: the integral that encodes binomial structure, linked to gamma by $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$.
- **Riemann zeta function** $\zeta(s)$ for $\operatorname{Re} s > 1$: series, integral representation, Euler product, and the Basel value $\zeta(2) = \pi^2/6$.

The common question is how to extend a discrete pattern to a function we can differentiate, integrate, and use in later chapters. Write $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

Block stacking and the harmonic numbers

Step 1: find the increment. How far can n identical blocks (length two units, unit mass) overhang a table edge? With one block we balance it half off, so its center of mass sits at the edge $x = 0$; the overhang is $H_1 = 1$.

Suppose $n - 1$ stacked blocks already have their center of mass at the edge, with total overhang H_{n-1} . Slip one more block underneath with its center at -1 , so its end lines up with the table edge. Each block has unit mass, so the new center of mass x of all n blocks is the mass-weighted average (total mass n times x equals the sum of mass times position): the top $n - 1$ blocks contribute mass $n - 1$ at position 0, and the new block mass 1 at position -1 , so

$$nx = (n - 1) \cdot 0 + 1 \cdot (-1) \implies x = -\frac{1}{n}.$$

Shifting the whole stack forward by $1/n$ returns its center of mass to the edge and increases the overhang by the same amount. Thus $H_n = H_{n-1} + 1/n$; starting from $H_0 = 0$, induction gives

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} = \sum_{k=1}^n \frac{1}{k}. \quad (2.1.1)$$

Block stacking and the harmonic numbers (cont.)

Step 2: name the sequence. The series $1 + \frac{1}{2} + \frac{1}{3} + \dots$ is the *harmonic series*; its partial sums (2.1.1) are the *harmonic numbers* $\{1, \frac{3}{2}, \frac{11}{6}, \frac{25}{12}, \dots\}$. They diverge, but extraordinarily slowly: the estimate derived below places the threshold $H_n > 100$ at roughly 1.5×10^{43} terms.

This raises two questions we now answer.

- What natural analytic function interpolates the discrete sequence H_n ?
- How slowly does the sequence diverge?

The harmonic series is the special case $p = 1$ of the p -series $\sum_{k=1}^{\infty} k^{-p}$, which reappears as the zeta function at the end of this chapter.

Interpolating the partial sums

A finite sum as an integral. The geometric identity gives

$$\frac{1 - x^n}{1 - x} = 1 + x + \cdots + x^{n-1},$$

with continuous value n at $x = 1$. Integrating each power on $[0, 1]$ gives

$$\int_0^1 \frac{1 - x^n}{1 - x} dx = \sum_{k=0}^{n-1} \frac{1}{k+1} = H_n.$$

This suggests the interpolation

$$f(s) = \int_0^1 \frac{1 - x^s}{1 - x} dx, \quad \operatorname{Re} s > -1, \quad (2.1.2)$$

where $x^s = e^{s \log x}$ uses the real logarithm on $0 < x < 1$. Near zero, $x^{\operatorname{Re} s}$ is integrable under the stated condition. At one, l'Hôpital's rule gives $(1 - x^s)/(1 - x) \rightarrow s$. Thus $f(n) = H_n$, and the same increment follows directly:

$$f(s+1) - f(s) = \int_0^1 x^s dx = \frac{1}{s+1}, \quad f(0) = 0.$$

Interpolating the partial sums (cont.)

The difference between a staircase and a logarithm. Because $1/x$ decreases, comparison with the area under the curve gives

$$H_n > \int_1^{n+1} \frac{dx}{x} = \log(n+1) > \log n.$$

The positive difference $a_n = H_n - \log n$ decreases, since

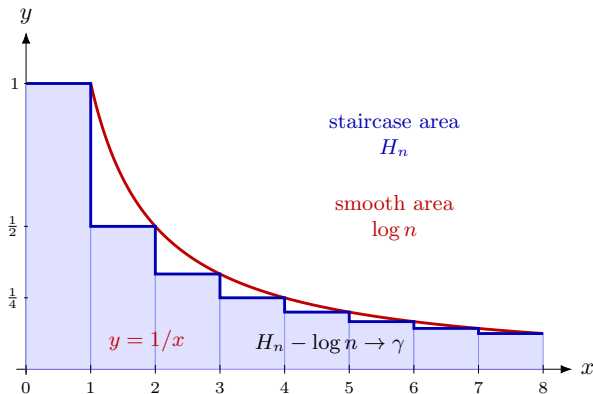
$$a_{n+1} - a_n = \frac{1}{n+1} - \int_n^{n+1} \frac{dx}{x} < 0.$$

A decreasing sequence bounded below has a limit. We call it the Euler–Mascheroni constant:

$$\gamma = \lim_{n \rightarrow \infty} (H_n - \log n) = 0.577215664901532 \dots \quad (2.1.3)$$

Equivalently, $H_n = \log n + \gamma + o(1)$, where $o(1)$ tends to zero. The next picture shows the two areas whose difference approaches γ .

Euler–Mascheroni as a staircase error



The harmonic number H_n is the area of a unit-width staircase. The logarithm $\log n = \int_1^n dx/x$ is the area under the smooth curve. The limiting excess area is $\gamma = \lim_{n \rightarrow \infty} (H_n - \log n)$.

Definition and the factorial recursion

Step 1: define the canonical interpolation. Infinitely many functions can agree with $n!$ at the nonnegative integers, so interpolation alone does not imply uniqueness. The standard canonical choice is the *gamma function*, defined initially by Euler's integral of the second kind,

$$\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt, \quad \operatorname{Re} z > 0. \quad (2.2.1)$$

Writing $\sigma = \operatorname{Re} z$, the endpoint tests are immediate: near $t = 0$, where $a(t) \sim b(t)$ means $a(t)/b(t) \rightarrow 1$, the integrand behaves like $t^{\sigma-1}$ and is integrable exactly when $\sigma > 0$; at infinity the exponential dominates every power. The integral therefore converges absolutely and defines a holomorphic function for $\operatorname{Re} z > 0$. Our main tool will be its integration-by-parts recursion.

Step 2: derive the recursion. Apply integration by parts to $\Gamma(z+1)$, taking

$$u = t^z, \quad dv = e^{-t} dt, \quad du = z t^{z-1} dt, \quad v = -e^{-t}.$$

Definition and the factorial recursion (cont.)

Then

$$\begin{aligned}\Gamma(z+1) &= \int_0^{\infty} e^{-t} t^z dt \\ &= \left[-t^z e^{-t} \right]_0^{\infty} + z \int_0^{\infty} e^{-t} t^{z-1} dt.\end{aligned}$$

For $\sigma > 0$,

$$|t^z e^{-t}| = t^{\sigma} e^{-t} \longrightarrow 0 \quad (t \rightarrow \infty), \quad |t^z e^{-t}| = t^{\sigma} e^{-t} \longrightarrow 0 \quad (t \rightarrow 0^+),$$

so the boundary term is zero. Therefore

$$\Gamma(z+1) = z \Gamma(z). \tag{2.2.2}$$

Step 3: recover factorials. At $z = 1$,

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = \left[-e^{-t} \right]_0^{\infty} = 1.$$

Definition and the factorial recursion (cont.)

For $n \in \mathbb{N}$, repeated use of the recursion gives

$$\begin{aligned}\Gamma(n+1) &= n\Gamma(n) = n(n-1)\Gamma(n-1) \\ &= \cdots = n(n-1)\cdots 2 \cdot 1 \Gamma(1) = n!.\end{aligned}$$

Together with $\Gamma(1) = 1 = 0!$, this proves

$$\Gamma(n+1) = n!, \quad n = 0, 1, 2, \dots \quad (2.2.3)$$

Keep track of the shift: $\Gamma(n+1) = n!$, not $\Gamma(n) = n!$.

A second integral form follows from $t = u^2$. Since $dt = 2u du$ and $t^{z-1} = (u^2)^{z-1} = u^{2z-2}$ for $u > 0$,

$$\begin{aligned}\Gamma(z) &= \int_0^\infty e^{-u^2} u^{2z-2} (2u) du \\ &= 2 \int_0^\infty e^{-u^2} u^{2z-1} du, \quad \operatorname{Re} z > 0.\end{aligned}$$

Hence

$$\Gamma(z) = 2 \int_0^\infty e^{-u^2} u^{2z-1} du. \quad (2.2.4)$$

Worked examples: half-integer values

Step 1: evaluate the Gaussian integral. First evaluate the Gaussian integral

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

The integrand is positive, so $I > 0$. Squaring and writing the product as a double integral (the integrand is nonnegative),

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) \\ &= \int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy. \end{aligned}$$

Worked examples: half-integer values (cont.)

The double integral adds contributions over the plane. In polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, a small area element has sides dr and $r d\theta$, hence $dx dy = r dr d\theta$. Thus

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^\infty e^{-r^2} r dr d\theta \\ &= \int_0^{2\pi} \left[-\frac{1}{2} e^{-r^2} \right]_{r=0}^{r=\infty} d\theta \\ &= \int_0^{2\pi} \frac{1}{2} d\theta = \pi. \end{aligned}$$

Because $I > 0$, $I = \sqrt{\pi}$.

Step 2: insert $z = \frac{1}{2}$. Set $z = \frac{1}{2}$ in the Gaussian form (2.2.4). Since $2z - 1 = 0$,

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= 2 \int_0^\infty e^{-u^2} du \\ &= \int_{-\infty}^\infty e^{-u^2} du = \sqrt{\pi}. \end{aligned}$$

Worked examples: half-integer values (cont.)

Thus

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}. \quad (2.2.5)$$

One application of the recursion gives

$$\begin{aligned} \Gamma\left(\frac{3}{2}\right) &= \left(\frac{3}{2} - 1\right) \Gamma\left(\frac{3}{2} - 1\right) \\ &= \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}. \end{aligned}$$

More generally, apply the recursion m times, exactly as in the $\Gamma(3/2)$ step above:

$$\Gamma\left(m + \frac{1}{2}\right) = \left(m - \frac{1}{2}\right) \left(m - \frac{3}{2}\right) \cdots \frac{1}{2} \Gamma\left(\frac{1}{2}\right).$$

Each of the m factors is $m - k + \frac{1}{2} = \frac{2(m-k)+1}{2}$ for $k = 1, \dots, m$, so their numerators run through the odd numbers $2m-1, 2m-3, \dots, 1$ and each contributes a $\frac{1}{2}$. With $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ this is

$$\Gamma\left(m + \frac{1}{2}\right) = \frac{1 \cdot 3 \cdots (2m-1)}{2^m} \sqrt{\pi}.$$

Worked examples: half-integer values (cont.)

To put it in closed form, separate the even and odd factors of $(2m)!$,

$$(2m)! = (2^m m!) [1 \cdot 3 \cdots (2m-1)] \implies 1 \cdot 3 \cdots (2m-1) = \frac{(2m)!}{2^m m!},$$

so that $\frac{1 \cdot 3 \cdots (2m-1)}{2^m} = \frac{(2m)!}{4^m m!}$ and

$$\Gamma\left(m + \frac{1}{2}\right) = \frac{(2m)!}{4^m m!} \sqrt{\pi}.$$

The formula also holds at $m = 0$, with the empty product equal to one.

Analytic continuation to the left half-plane

The defining integral (2.2.1) converges only for $\operatorname{Re} z > 0$, but the recursion can be solved backward:

$$\Gamma(z) = \frac{\Gamma(z+1)}{z} = \frac{1}{z} \int_0^\infty e^{-t} t^z dt, \quad \operatorname{Re} z > -1, \quad z \neq 0.$$

Repeating the recursion m times gives

$$\begin{aligned} \Gamma(z+m) &= (z+m-1)\Gamma(z+m-1) \\ &= (z+m-1)(z+m-2)\Gamma(z+m-2) \\ &= \cdots = (z+m-1)(z+m-2)\cdots(z+1)z\Gamma(z). \end{aligned}$$

If m is chosen so that $\operatorname{Re}(z+m) > 0$, then $\Gamma(z+m)$ is defined by the original integral and

$$\Gamma(z) = \frac{\Gamma(z+m)}{z(z+1)(z+2)\cdots(z+m-1)}. \quad (2.2.6)$$

This gives a meromorphic continuation to the entire complex plane, independent of the admissible choice of m .

Analytic continuation to the left half-plane (cont.)

To see the nature of the singularity at $z = -n$, where $n \in \mathbb{N}_0$, set $z = -n + \varepsilon$ and choose $m = n + 1$:

$$\Gamma(-n + \varepsilon) = \frac{\Gamma(1 + \varepsilon)}{(-n + \varepsilon)(1 - n + \varepsilon) \cdots (-1 + \varepsilon)\varepsilon}.$$

As $\varepsilon \rightarrow 0$,

$$\Gamma(1 + \varepsilon) \rightarrow \Gamma(1) = 1$$

and, with the empty product interpreted as 1 when $n = 0$,

$$(-n)(1 - n) \cdots (-1) = (-1)^n n!.$$

Therefore

$$\Gamma(-n + \varepsilon) = \frac{(-1)^n}{n! \varepsilon} + O(1),$$

where $F = O(G)$ means that $|F/G|$ remains bounded in the stated limit. Thus every non-positive integer is a simple pole and

$$\operatorname{Res}_{z=-n} \Gamma(z) = \frac{(-1)^n}{n!}. \quad (2.2.7)$$

Analytic continuation to the left half-plane (cont.)

A more efficient relation between positive and negative arguments is Euler's reflection formula.

Euler's reflection formula

For $z \notin \mathbb{Z}$,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

First take real $0 < a < 1$. Both gamma integrals then converge absolutely:

$$\begin{aligned}\Gamma(a)\Gamma(1-a) &= \left(\int_0^\infty t^{a-1} e^{-t} dt \right) \left(\int_0^\infty s^{-a} e^{-s} ds \right) \\ &= \int_0^\infty \int_0^\infty t^{a-1} s^{-a} e^{-(t+s)} dt ds.\end{aligned}$$

Absolute convergence allows us to change the integration order (Fubini's theorem). For each fixed $s > 0$, set $t = su$. Then $dt = s du$, and

$$t^{a-1} s^{-a} dt = (su)^{a-1} s^{-a} (s du) = u^{a-1} du.$$

Analytic continuation to the left half-plane (cont.)

Consequently

$$\begin{aligned}\Gamma(a)\Gamma(1-a) &= \int_0^\infty \int_0^\infty u^{a-1} e^{-s(1+u)} du ds \\ &= \int_0^\infty u^{a-1} \left(\int_0^\infty e^{-s(1+u)} ds \right) du \\ &= \int_0^\infty u^{a-1} \left[\frac{-e^{-s(1+u)}}{1+u} \right]_{s=0}^{s=\infty} du \\ &= \int_0^\infty \frac{u^{a-1}}{1+u} du.\end{aligned}$$

Chapter 1 already evaluated the last integral by a keyhole contour:

$$\int_0^\infty \frac{u^{a-1}}{1+u} du = \frac{\pi}{\sin(\pi a)}, \quad 0 < a < 1.$$

Thus $\Gamma(a)\Gamma(1-a) = \pi/\sin(\pi a)$. Analytic continuation extends the identity to every noninteger complex argument. There is no need to repeat the contour calculation here.

Analytic continuation to the left half-plane (cont.)

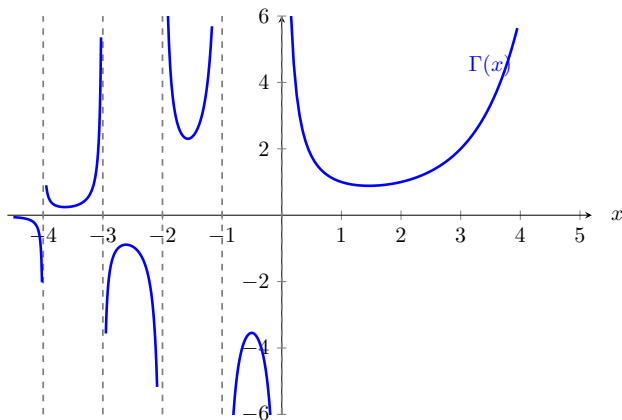
A useful consequence: Γ has no zeros. At a noninteger argument the reflection product is nonzero. At positive integers Γ is a factorial; the remaining integers are poles, not zeros.

Replacing the reflection argument by $-z$ gives the equivalent form

$$\Gamma(-z) = -\frac{\pi}{\Gamma(z+1)\sin(\pi z)}, \quad z \notin \mathbb{Z}. \quad (2.2.8)$$

As a function of z , $\Gamma(-z)$ has poles at $z = 0, 1, 2, \dots$; equivalently, $\Gamma(w)$ has poles at $w = 0, -1, -2, \dots$

Real-axis picture of Γ



On the positive real axis, $\Gamma(n+1) = n!$. Because $\Gamma(z) = \Gamma(z+n+1)/(z(z+1)\cdots(z+n))$, running the recursion backward makes the factor $z+n$ produce a simple pole at each $z = -n$, with residue $(-1)^n/n!$; the alternating sign of that residue is why Γ flips sign from one pole-interval to the next.

Euler's limit representation

Euler's integral leads to a finite-product formula when we replace its exponential by an elementary limit. For $\operatorname{Re} z > 0$, define

$$F_n(z) = \int_0^n \left(1 - \frac{t}{n}\right)^n t^{z-1} dt.$$

For each fixed $t \geq 0$, the integrand (extended by zero for $t > n$) tends to $e^{-t} t^{z-1}$. Moreover, $\log(1-u) \leq -u$ for $0 \leq u < 1$, so

$$\left| \left(1 - \frac{t}{n}\right)^n t^{z-1} \right| \leq e^{-t} t^{\operatorname{Re} z - 1}, \quad 0 < t < n.$$

The right side is integrable when $\operatorname{Re} z > 0$. Dominated convergence therefore gives

$$\lim_{n \rightarrow \infty} F_n(z) = \int_0^\infty e^{-t} t^{z-1} dt = \Gamma(z).$$

Now put $t = nu$. Since $dt = n du$ and $(nu)^{z-1} = n^{z-1} u^{z-1}$,

$$F_n(z) = n^z \int_0^1 (1-u)^n u^{z-1} du.$$

Euler's limit representation (cont.)

Integrate by parts, using $u^{z-1} du = d(u^z)/z$. The endpoint term vanishes, and hence

$$\int_0^1 (1-u)^n u^{z-1} du = \frac{n}{z} \int_0^1 (1-u)^{n-1} u^z du.$$

Repeating this step n times gives

$$\begin{aligned} \int_0^1 (1-u)^n u^{z-1} du &= \frac{n!}{z(z+1)\cdots(z+n-1)} \int_0^1 u^{z+n-1} du \\ &= \frac{n!}{z(z+1)\cdots(z+n)}. \end{aligned}$$

Combining this identity with the limit of F_n proves

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{n^z n!}{z(z+1)\cdots(z+n)}, \quad \operatorname{Re} z > 0. \quad (2.2.9)$$

The recursion extends the formula to every $z \notin \{0, -1, -2, \dots\}$. Indeed, choose an integer $m \geq 0$ with $\operatorname{Re}(z+m) > 0$, apply the formula to $\Gamma(z+m)$, and divide by

Euler's limit representation (cont.)

$z(z+1)\cdots(z+m-1)$. If $A_n(z)$ denotes the approximant in (2.2.9), the resulting approximant is

$$\frac{n^{z+m}n!}{z(z+1)\cdots(z+n+m)} = A_n(z) \prod_{j=1}^m \frac{n}{n+z+j}.$$

The finite product tends to 1. Thus the same displayed limit holds away from the poles.

Hankel's contour formula for $1/\Gamma$

Chapter 1's keyhole contour turned a multivalued power into a computable integral. The same move gives a representation of $1/\Gamma$ that works where the defining integral (2.2.1) does not.

Step 1: the Hankel contour. Cut the plane along the negative real axis and let $\mathcal{H}$ start at $-\infty$ *below* the cut, circle the origin counterclockwise, and return to $-\infty$ *above* the cut. Use $t^{-a} = e^{-a \log t}$ with $-\pi < \arg t < \pi$. Since the integrand $e^t t^{-a}$ decays like $e^{-|t|}$ on both legs, the integral converges for every complex a and defines an entire function of a .

Step 2: collapse onto the cut for $0 < a < 1$. Shrink the circular part to zero. On the lower leg $t = re^{-i\pi}$ with $r : \infty \rightarrow 0$, so $dt = -dr$ and $t^{-a} = r^{-a} e^{i\pi a}$. On the upper leg $t = re^{i\pi}$ with $r : 0 \rightarrow \infty$ so again $dt = -dr$, while $t^{-a} = r^{-a} e^{-i\pi a}$. Thus, before taking $\varepsilon \rightarrow 0$, the two straight pieces contribute

$$\int_{\text{lower}} e^t t^{-a} dt = e^{i\pi a} \int_{\varepsilon}^{\infty} e^{-r} r^{-a} dr, \quad \int_{\text{upper}} e^t t^{-a} dt = -e^{-i\pi a} \int_{\varepsilon}^{\infty} e^{-r} r^{-a} dr.$$

The circle contributes $O(\varepsilon^{1-a}) \rightarrow 0$. Combining the two legs,

$$\int_{\mathcal{H}} e^t t^{-a} dt = (e^{i\pi a} - e^{-i\pi a}) \int_0^{\infty} e^{-r} r^{-a} dr = 2i \sin(\pi a) \Gamma(1-a).$$

Hankel's contour formula for $1/\Gamma$ (cont.)

Euler's reflection formula, $\sin(\pi a)\Gamma(1-a) = \pi/\Gamma(a)$, then gives

$$\frac{1}{2\pi i} \int_{\mathcal{H}} e^t t^{-a} dt = \frac{1}{\Gamma(a)}. \quad (2.2.10)$$

Step 3: extend to every a . The left side is entire in a , and the right side is entire because Γ never vanishes: at a noninteger the reflection product is nonzero, at positive integers Γ is a factorial, and at the remaining integers Γ has poles, not zeros. The two sides agree on $(0, 1)$, so the identity theorem of Chapter 1 extends (2.2.10) to all complex a .

Why this contour matters. An open contour keeps a multivalued power legal where a closed loop cannot. In Chapter 5 the same $\mathcal{H}$ turns a noninteger-order Bessel integral into its power series one Gamma factor at a time—the Sommerfeld–Watson reduction used in scattering theory.

Summary of gamma formulas

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt, \quad \operatorname{Re} z > 0,$$

$$\Gamma(z+1) = z\Gamma(z),$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi},$$

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}, \quad z \notin \mathbb{Z},$$

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{n^z n!}{z(z+1) \cdots (z+n)}, \quad z \notin \{0, -1, -2, \dots\},$$

$$\frac{1}{\Gamma(z)} = \frac{1}{2\pi i} \int_{\mathcal{H}} e^t t^{-z} dt, \quad \mathcal{H}: \text{Hankel contour (2.2.10).}$$

For $n \in \mathbb{N}_0$, $\Gamma(n+1) = n!$. The meromorphic continuation has simple poles at $0, -1, -2, \dots$, with residue $(-1)^n/n!$ at $z = -n$.

Worked examples: gamma integrals

Example 1. Evaluate

$$\int_0^{\infty} e^{-x^p} dx, \quad p > 0.$$

Set $t = x^p$. Then

$$x = t^{1/p}, \quad dx = \frac{1}{p} t^{1/p-1} dt,$$

and the endpoints $x = 0, \infty$ map to $t = 0, \infty$. Therefore

$$\begin{aligned} \int_0^{\infty} e^{-x^p} dx &= \frac{1}{p} \int_0^{\infty} e^{-t} t^{1/p-1} dt \\ &= \frac{1}{p} \Gamma\left(\frac{1}{p}\right). \end{aligned}$$

Using

$$\Gamma\left(1 + \frac{1}{p}\right) = \frac{1}{p} \Gamma\left(\frac{1}{p}\right),$$

we obtain

$$\int_0^{\infty} e^{-x^p} dx = \Gamma\left(1 + \frac{1}{p}\right). \quad (2.2.11)$$

Worked examples: gamma integrals (cont.)

Example 2. For $p > 1$, evaluate $\int_0^\infty \cos(x^p) dx$. Write the cosine integral as the real part of $\int_0^\infty e^{ix^p} dx$. Use a sector between the positive real axis and angle $\pi/(2p)$, with radii ε and R . For noninteger p , choose $z^p = e^{p \operatorname{Log} z}$; the sector stays away from the negative-real-axis cut. The small arc avoids the branch point at zero. On the slanted ray $z = re^{i\pi/(2p)}$,

$$z^p = r^p e^{i\pi/2} = ir^p, \quad iz^p = i \cdot ir^p = -r^p, \quad dz = e^{i\pi/(2p)} dr,$$

so the outward ray integral is

$$\int_{\text{ray}} e^{iz^p} dz = e^{i\pi/(2p)} \int_0^\infty e^{-r^p} dr.$$

The real integral is the one evaluated in Example 1:

$$\int_0^\infty e^{-r^p} dr = \Gamma\left(1 + \frac{1}{p}\right).$$

Worked examples: gamma integrals (cont.)

On the outer arc, $|e^{iz^p}| = e^{-R^p \sin(p\theta)}$. With $\varphi = p\theta$ and $\sin \varphi \geq 2\varphi/\pi$ on $[0, \pi/2]$,

$$\begin{aligned} \left| \int_{\text{outer arc}} e^{iz^p} dz \right| &\leq \frac{R}{p} \int_0^{\pi/2} e^{-R^p \sin \varphi} d\varphi \\ &\leq \frac{R}{p} \int_0^{\pi/2} e^{-2R^p \varphi/\pi} d\varphi \\ &= \frac{\pi}{2p} R^{1-p} (1 - e^{-R^p}) \rightarrow 0. \end{aligned}$$

This is where $p > 1$ is needed; at $p = 1$, $\int_0^\infty \cos x \, dx$ does not converge as an ordinary improper integral. The inner arc vanishes as $\varepsilon \rightarrow 0$ by length $\rightarrow 0$. The positive-real segment is traversed from ε to R , whereas the slanted segment is traversed from R back to ε . Cauchy's theorem therefore gives the oriented identity

$$\int_\varepsilon^R e^{ix^p} dx + \int_{\text{outer}} e^{iz^p} dz + e^{i\pi/(2p)} \int_R^\varepsilon e^{-r^p} dr + \int_{\text{inner}} e^{iz^p} dz = 0.$$

Worked examples: gamma integrals (cont.)

Letting $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$ makes both arc integrals vanish and reverses the limits on the slanted-ray integral, so

$$\int_0^\infty e^{ix^p} dx = e^{i\pi/(2p)} \frac{1}{p} \Gamma\left(\frac{1}{p}\right) = e^{i\pi/(2p)} \Gamma\left(1 + \frac{1}{p}\right).$$

Taking the real part yields

$$\int_0^\infty \cos(x^p) dx = \frac{1}{p} \Gamma\left(\frac{1}{p}\right) \cos\left(\frac{\pi}{2p}\right) = \Gamma\left(1 + \frac{1}{p}\right) \cos\left(\frac{\pi}{2p}\right), \quad p > 1. \quad (2.2.12)$$

Stirling's approximation

This is Laplace's method (Chapter 1): a large-parameter integral is dominated by a neighborhood of the maximum of the real phase, and that neighborhood contributes a Gaussian. Start from the gamma integral

$$n! = \Gamma(n+1) = \int_0^{\infty} t^n e^{-t} dt.$$

Set $t = nx$, so $dt = n dx$. Then

$$\begin{aligned} n! &= \int_0^{\infty} (nx)^n e^{-nx} n dx \\ &= n^{n+1} \int_0^{\infty} e^{n \log x - nx} dx \\ &= n^{n+1} e^{-n} \int_0^{\infty} e^{n(\log x - x + 1)} dx. \end{aligned}$$

Define

$$h(x) = \log x - x + 1.$$

Its derivatives are

$$h'(x) = \frac{1}{x} - 1, \quad h''(x) = -\frac{1}{x^2} < 0.$$

Stirling's approximation (cont.)

Thus h is strictly concave, has its unique maximum at $x = 1$, and

$$h(1) = 0, \quad h''(1) = -1.$$

The mass of the integral is therefore concentrated near $x = 1$.

Localize near the maximum. Taylor's theorem with $h(1) = 0$, $h'(1) = 0$, $h''(1) = -1$ gives $n h(x) \approx -\frac{n}{2}(x-1)^2$ near the maximum, so $e^{nh(x)}$ stays of order one only while $(x-1)^2 \lesssim 1/n$, that is $|x-1| \sim n^{-1/2}$. This fixes the natural width of the peak; rescaling it to order one motivates writing

$$x = 1 + \frac{u}{\sqrt{n}}, \quad dx = \frac{du}{\sqrt{n}}.$$

As x runs from 0 to ∞ , u runs from $-\sqrt{n}$ to ∞ . Hence

$$\int_0^\infty e^{nh(x)} dx = \frac{1}{\sqrt{n}} \int_{-\sqrt{n}}^\infty \exp \left[n \left(\log \left(1 + \frac{u}{\sqrt{n}} \right) - \frac{u}{\sqrt{n}} \right) \right] du.$$

Only the quadratic Taylor term is needed for the leading approximation:

$$h \left(1 + \frac{u}{\sqrt{n}} \right) = -\frac{u^2}{2n} + O(n^{-3/2}) \quad \text{for fixed } u.$$

Stirling's approximation (cont.)

Multiplying by n makes the exponent tend to $-u^2/2$.

Reduce to a Gaussian. Away from $x = 1$, $h(x) < 0$, so $e^{nh(x)}$ becomes exponentially small. The leading contribution is the quadratic peak just calculated. Extending its rescaled limits gives

$$\int_0^\infty e^{nh(x)} dx \sim \frac{1}{\sqrt{n}} \int_{-\infty}^\infty e^{-u^2/2} du = \sqrt{\frac{2\pi}{n}}.$$

This is the local-Gaussian approximation of Chapter 1.

Substitute this into the scaled gamma integral:

$$n! \sim n^{n+1} e^{-n} \sqrt{\frac{2\pi}{n}} = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n. \quad (2.2.13)$$

Taking logarithms gives the corresponding leading expansion

$$\log(n!) = n \log n - n + \frac{1}{2} \log(2\pi n) + o(1).$$

Definition and trigonometric form

The *Euler beta function* is

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt, \quad \operatorname{Re} p > 0, \quad \operatorname{Re} q > 0. \quad (2.3.1)$$

Near 0, the magnitude behaves as $t^{\operatorname{Re} p-1}$; near 1, it behaves as $(1-t)^{\operatorname{Re} q-1}$. The two restrictions are therefore just the usual power-integral tests at the two endpoints.

The function is symmetric: the substitution $u = 1 - t$ gives $B(q, p) = B(p, q)$.

For nonnegative integers m, n , the integral reduces to a factorial ratio:

$$B(m+1, n+1) = \frac{m! n!}{(m+n+1)!} = \frac{1}{(m+n+1) \binom{m+n}{m}}. \quad (2.3.2)$$

This follows directly from the Beta–Gamma identity proved next, using $\Gamma(k+1) = k!$.

We do not need a separate repeated-integration-by-parts proof.

For the trigonometric form, set

$$t = \sin^2 \theta, \quad dt = 2 \sin \theta \cos \theta d\theta.$$

As t runs from 0 to 1, θ runs from 0 to $\pi/2$, and

$$1 - t = 1 - \sin^2 \theta = \cos^2 \theta.$$

Definition and trigonometric form (cont.)

Therefore

$$\begin{aligned}t^{p-1} &= (\sin^2 \theta)^{p-1} = \sin^{2p-2} \theta, \\(1-t)^{q-1} &= (\cos^2 \theta)^{q-1} = \cos^{2q-2} \theta.\end{aligned}$$

Substitution into (2.3.1) yields

$$\begin{aligned}B(p, q) &= \int_0^{\pi/2} \sin^{2p-2} \theta \cos^{2q-2} \theta \, 2 \sin \theta \cos \theta \, d\theta \\&= 2 \int_0^{\pi/2} \sin^{2p-1} \theta \cos^{2q-1} \theta \, d\theta.\end{aligned}\tag{2.3.3}$$

Beta in terms of gamma

Beta–gamma identity

For $\operatorname{Re} p > 0$ and $\operatorname{Re} q > 0$,

$$B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

Use the Gaussian gamma representation (2.2.4):

$$\Gamma(p) = 2 \int_0^\infty e^{-x^2} x^{2p-1} dx, \quad \Gamma(q) = 2 \int_0^\infty e^{-y^2} y^{2q-1} dy.$$

Because the product is absolutely integrable for the stated parameter range, Fubini's theorem gives

$$\Gamma(p)\Gamma(q) = 4 \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2p-1} y^{2q-1} dx dy.$$

In the first quadrant, set

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r \geq 0, \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

Beta in terms of gamma (cont.)

The polar area element is $dx dy = r dr d\theta$, as in the Gaussian integral. Keep this extra factor of r when combining powers. Combining the radial powers, including the area factor, gives $r^{2p-1} r^{2q-1} r = r^{2(p+q)-1}$. Thus

$$\Gamma(p)\Gamma(q) = 4 \left(\int_0^\infty e^{-r^2} r^{2(p+q)-1} dr \right) \left(\int_0^{\pi/2} \cos^{2p-1} \theta \sin^{2q-1} \theta d\theta \right).$$

Split $4 = 2 \cdot 2$. The radial factor is

$$2 \int_0^\infty e^{-r^2} r^{2(p+q)-1} dr = \Gamma(p+q).$$

The angular factor is, by (2.3.3),

$$2 \int_0^{\pi/2} \sin^{2q-1} \theta \cos^{2p-1} \theta d\theta = B(q, p).$$

Since $B(q, p) = B(p, q)$,

$$\Gamma(p)\Gamma(q) = \Gamma(p+q)B(p, q).$$

Since Γ has no zeros and is finite here, division by $\Gamma(p+q)$ gives

$$B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}. \quad (2.3.4)$$

Summary of beta formulas

$$B(p, q) = \frac{\Gamma(p) \Gamma(q)}{\Gamma(p + q)},$$

$$B(m + 1, n + 1) = \frac{m! n!}{(m + n + 1)!} = \frac{1}{(m + n + 1) \binom{m+n}{m}},$$

$$B(p, q) = \int_0^1 t^{p-1} (1 - t)^{q-1} dt, \quad \operatorname{Re} p, \operatorname{Re} q > 0,$$

$$B(p, q) = 2 \int_0^{\pi/2} \sin^{2p-1} \theta \cos^{2q-1} \theta d\theta.$$

The first line lets us evaluate many integrals in closed form once we recognize their $t^{p-1}(1-t)^{q-1}$ shape.

Worked examples: beta integrals

Example 1. Evaluate

$$I = \int_0^1 \frac{dx}{\sqrt{1-x^4}}.$$

Set $u = x^4$. Then

$$x = u^{1/4}, \quad dx = \frac{1}{4} u^{-3/4} du,$$

and the limits remain 0 and 1. Hence

$$\begin{aligned} I &= \frac{1}{4} \int_0^1 u^{-3/4} (1-u)^{-1/2} du \\ &= \frac{1}{4} B\left(\frac{1}{4}, \frac{1}{2}\right) \\ &= \frac{1}{4} \frac{\Gamma(1/4)\Gamma(1/2)}{\Gamma(3/4)} \\ &= \frac{\sqrt{\pi} \Gamma(1/4)}{4\Gamma(3/4)}. \end{aligned}$$

The answer is therefore

$$I = \frac{\sqrt{\pi} \Gamma(1/4)}{4\Gamma(3/4)}. \quad (2.3.5)$$

Worked examples: beta integrals (cont.)

Example 2. Evaluate

$$I_n = \int_0^{\pi} \sin^n x \, dx, \quad n > -1.$$

The condition $n > -1$ ensures integrability at 0 and π . Since $\sin(\pi - x) = \sin x$,

$$I_n = 2 \int_0^{\pi/2} \sin^n x \, dx.$$

Compare this with

$$B(p, q) = 2 \int_0^{\pi/2} \sin^{2p-1} x \cos^{2q-1} x \, dx.$$

To obtain $\sin^n x$ with no cosine factor, solve

$$2p - 1 = n, \quad 2q - 1 = 0 \quad \implies \quad p = \frac{n+1}{2}, \quad q = \frac{1}{2}.$$

Use $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ and $\Gamma(1/2) = \sqrt{\pi}$:

$$I_n = B\left(\frac{n+1}{2}, \frac{1}{2}\right) = \frac{\Gamma((n+1)/2)\Gamma(1/2)}{\Gamma((n+2)/2)} = \frac{\sqrt{\pi} \Gamma((n+1)/2)}{\Gamma((n+2)/2)}. \quad (2.3.6)$$

From the p -series to $\zeta(s)$

Step 1: define ζ and check convergence. For $s \in \mathbb{C}$ with $\sigma = \operatorname{Re} s > 1$, define

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad n^{-s} = e^{-s \log n}. \quad (2.4.1)$$

The series converges absolutely for $\sigma = \operatorname{Re} s > 1$ by the integral test, and diverges for $\sigma \leq 1$; at $s = 1$ it is the divergent harmonic series.

Step 2: convert the series into an integral. Weighting by t^{s-1} and integrating over $t \in (0, \infty)$ produces

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{t^{s-1}}{e^t - 1} dt, \quad \operatorname{Re} s > 1. \quad (2.4.2)$$

The same condition is visible near zero: $e^t - 1 \sim t$, so the integrand behaves like t^{s-2} and is integrable when $\operatorname{Re} s > 1$. At infinity the exponential denominator ensures convergence.

From the p -series to $\zeta(s)$ (cont.)

Step 3: expand and integrate term by term. To prove (2.4.2), use the geometric series. For $t > 0$,

$$\begin{aligned}\frac{1}{e^t - 1} &= \frac{e^{-t}}{1 - e^{-t}} \\ &= e^{-t} \sum_{k=0}^{\infty} e^{-kt} \\ &= \sum_{n=1}^{\infty} e^{-nt}.\end{aligned}$$

For $\sigma = \operatorname{Re} s > 1$, term-by-term integration is valid because $\sum_{n \geq 1} \int_0^{\infty} t^{\sigma-1} e^{-nt} dt = \Gamma(\sigma) \sum_{n \geq 1} n^{-\sigma} < \infty$. Hence

$$\int_0^{\infty} \frac{t^{s-1}}{e^t - 1} dt = \sum_{n=1}^{\infty} \int_0^{\infty} t^{s-1} e^{-nt} dt.$$

From the p -series to $\zeta(s)$ (cont.)

In each integral put $u = nt$, so $t = u/n$ and $dt = du/n$:

$$\begin{aligned}\int_0^\infty t^{s-1} e^{-nt} dt &= \int_0^\infty \left(\frac{u}{n}\right)^{s-1} e^{-u} \frac{du}{n} \\ &= n^{-s} \int_0^\infty u^{s-1} e^{-u} du \\ &= n^{-s} \Gamma(s).\end{aligned}$$

Hence

$$\begin{aligned}\int_0^\infty \frac{t^{s-1}}{e^t - 1} dt &= \Gamma(s) \sum_{n=1}^\infty n^{-s} \\ &= \Gamma(s) \zeta(s).\end{aligned}$$

Division by $\Gamma(s)$ proves (2.4.2).

The Euler product over primes

Let $s \in \mathbb{C}$ with $\sigma = \operatorname{Re} s > 1$. Following the book, remove from the Dirichlet series one prime factor at a time. Start with

$$\zeta(s) = 1 + 2^{-s} + 3^{-s} + 4^{-s} + 5^{-s} + \cdots.$$

Multiplying by 2^{-s} and subtracting removes precisely the terms whose indices are divisible by 2:

$$(1 - 2^{-s})\zeta(s) = 1 + 3^{-s} + 5^{-s} + 7^{-s} + \cdots.$$

Now multiply this remaining series by 3^{-s} and subtract again. This removes the indices divisible by 3 without restoring any even index:

$$(1 - 3^{-s})(1 - 2^{-s})\zeta(s) = 1 + 5^{-s} + 7^{-s} + 11^{-s} + 13^{-s} + \cdots.$$

After sieving by every prime $p \leq P$,

$$\prod_{p \leq P} (1 - p^{-s}) \zeta(s) = \sum_{\substack{n \geq 1 \\ \text{every prime } p|n \text{ satisfies } p > P}} n^{-s}.$$

Unique factorization explains the sieve: every integer $n > 1$ has a prime divisor, so it disappears as soon as that prime is included.

The Euler product over primes (cont.)

The passage to all primes is justified by absolute convergence. If a surviving integer is larger than 1, all of its prime divisors exceed P , and hence $n > P$. Therefore the magnitude of the surviving tail is at most

$$\sum_{n>P} n^{-\sigma} \rightarrow 0.$$

Only the term 1 remains, so

$$\left(\prod_{p \text{ prime}} (1 - p^{-s}) \right) \zeta(s) = 1.$$

Equivalently,

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}, \quad \operatorname{Re} s > 1. \quad (2.4.3)$$

A reference value. The classical Basel identity is used without proof:

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The Euler product over primes (cont.)

We do *not* develop meromorphic continuation of ζ in this chapter. The main point is that gamma supplies the weight in (2.4.2), so ζ lives in the same integral language as Γ and B .

For this course, the useful connection is between a sum of powers and a Gamma-weighted integral, not the theory of zeta zeros.

Where we have been

- **Gamma:** $\Gamma(n+1) = n!$, $\Gamma(z+1) = z\Gamma(z)$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, and $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ for $z \notin \mathbb{Z}$; poles at $0, -1, -2, \dots$ with residue $(-1)^n/n!$; Euler limit and Stirling $n! \sim \sqrt{2\pi n}(n/e)^n$; Hankel's contour (2.2.10) writes the entire function $1/\Gamma$ as an integral on $\mathcal{H}$.

- **Beta:**

$$B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} = \int_0^1 t^{p-1}(1-t)^{q-1} dt, \quad \operatorname{Re} p, \operatorname{Re} q > 0.$$

- **Zeta:**

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1} = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{t^{s-1}}{e^t - 1} dt, \quad \operatorname{Re} s > 1,$$

with the classical value $\zeta(2) = \pi^2/6$ quoted (not re-derived here).

Gamma and beta are the core language; later special functions are largely built from them.